SOC-Based Storage Participation

From the paper's compact model to a multi-agent complementarity problem

Where we are · September 2026

Let the operator price the state, not the participant

Storage has no marginal cost — only an opportunity cost, and that cost depends on the whole horizon, on the prices inside it, and on the state-of-charge trajectory. Baldick's proposal is to let the ISO carry the state of charge as a decision variable of its own clearing problem. The participant then submits only physical parameters plus a single scalar value function w(e) for the energy left at the end of the horizon. **The intertemporal opportunity cost is therefore computed by the operator, and no participant has to forecast prices in order to bid.**

Our question: is the resulting dispatch a competitive equilibrium?

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The models

Four statements of the same market, each one a step less compressed

As published: compact form

SOURCE PROBLEM S(9)

$$\min_{x,p,g,s} \left\{ \Delta T\, c^{\top} x + \Delta T\big(\epsilon \mathbf{1}^{\top} p + \zeta \mathbf{1}^{\top} g\big) - b\big(s_T - \bar{s}_T\big) \;\middle|\; \mathrm{S}(1)\text{–}\mathrm{S}(7),\quad \underbrace{p - g = Bx}_{\text{injection map}},\quad \underbrace{Cx \le d}_{\text{the whole rest of the market}} \right\}$$

I. Objective:
   I. First ($\Delta T\ c^T x$) is a vector of costs and offers and their respective offers/bids
   II. Second term $\Delta T(\epsilon 1^T p\ +\ \zeta 1^T g)$ is another vector that represents the degradation of the storage while charging/discharging
   III. The third term $b(s_T\ -\ \overline{s_T})$ is the willingness to pay of the storage based on future expectations opportunity costs

II. Constraints:
   I. Constraints $S(1) - S(7)$ represents all the storage constraints (charge/discharge limits, SOC boundaries, SOC energy balance)
   II. Everything that is not storage (network, generators, loads) is compressed into an injection map and a single polyhedron.
   III. Cx ≤ d is a single matrix inequality whose rows mix a generator's ramp limit, a line's thermal limit and a load's bid block, with no record of which is which.
   IV. A multi-agent model is built by splitting constraints along ownership lines. One cannot partition what has been collapsed.

So the first thing we had to do was undo the compression — not for presentation, but because the argument does not exist without it.

The expanded ISO clearing problem

OBJECTIVE

$$\min_{q,d,f,\theta,p,g,s} \quad \Delta T \sum_{t\in\mathcal{T}} \Big[\sum_{i,k} c_{i,k} q_{i,k,t} - \sum_{m,j} u_{m,j} d_{m,j,t} + \sum_{e} \big(\epsilon_e p_{e,t} + \zeta_e g_{e,t} \big) \Big] - \sum_{e} b_e \big(s_{e,T} - \bar{s}_{e,T} \big)$$

CONSTRAINTS · WITH THEIR MULTIPLIERS

$$\sum_{m\in\mathcal{D}(n),j} d_{m,j,t} + D_{n,t} + \sum_{\ell} A_{n\ell} f_{\ell,t} - \sum_{i\in\mathcal{G}(n),k} q_{i,k,t} - \sum_{e\in\mathcal{E}(n)} \big(g_{e,t} - p_{e,t} \big) = 0 \qquad \Delta T \lambda_{n,t} \text{ free} \qquad \text{(A1)}$$

$$y_\ell \big(\theta_{\mathrm{fr}(\ell),t} - \theta_{\mathrm{to}(\ell),t} \big) - f_{\ell,t} = 0 \qquad \Delta T \kappa_{\ell,t} \text{ free} \qquad \text{(A2)}$$

$$f_{\ell,t} - \bar{f}_\ell \le 0, \qquad -\bar{f}_\ell - f_{\ell,t} \le 0 \qquad \Delta T \bar{\mu}_{\ell,t},\ \Delta T \underline{\mu}_{\ell,t} \ge 0 \qquad \text{(A3)}$$

$$\theta_{r,t} = 0 \qquad \Delta T \xi_t \text{ free} \qquad \text{(A4)}$$

$$-q_{i,k,t} \le 0, \qquad q_{i,k,t} - \bar{q}_{i,k} \le 0 \qquad \Delta T \underline{\delta}_{i,k,t},\ \Delta T \bar{\delta}_{i,k,t} \ge 0 \qquad \text{(A5)}$$

$$Q_{i,t} - Q_{i,t-1} - \bar{R}_i \le 0, \qquad Q_{i,t-1} - Q_{i,t} - \underline{R}_i \le 0 \qquad \Delta T \bar{\tau}_{i,t},\ \Delta T \underline{\tau}_{i,t} \ge 0 \qquad \text{(A6)}$$

$$-d_{m,j,t} \le 0, \qquad d_{m,j,t} - \bar{d}_{m,j} \le 0 \qquad \Delta T \underline{\psi}_{m,j,t},\ \Delta T \bar{\psi}_{m,j,t} \ge 0 \qquad \text{(A7)}$$

$$s_{e,t} - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} + \Delta T \beta_e g_{e,t} = 0 \qquad \pi_{e,t} \text{ free} \qquad \text{(A8)}$$

$$\gamma_e s_{e,t-1} + \Delta T \alpha_e p_{e,t} - \bar{s}_e \le 0 \qquad \bar{\nu}_{e,t} \ge 0 \qquad \text{(A9)}$$

$$\underline{s}_e - \gamma_e s_{e,t-1} + \Delta T \beta_e g_{e,t} \le 0 \qquad \underline{\nu}_{e,t} \ge 0 \qquad \text{(A10)}$$

$$-p_{e,t} \le 0, \qquad p_{e,t} - \bar{p}_e \le 0 \qquad \Delta T \underline{\rho}_{e,t},\ \Delta T \bar{\rho}_{e,t} \ge 0 \qquad \text{(A11)}$$

$$-g_{e,t} \le 0, \qquad g_{e,t} - \bar{g}_e \le 0 \qquad \Delta T \underline{\sigma}_{e,t},\ \Delta T \bar{\sigma}_{e,t} \ge 0 \qquad \text{(A12)}$$

All constraints affine ⇒ a constraint qualification holds automatically. Power constraints carry ΔT on their multiplier; the energy constraints (A8)–(A10) do not.

Its KKT system — stationarity, one row per variable

(KA1)–(KA8)

$$
\begin{aligned}
q_{i,k,t}: \quad & \lambda_{n(i),t} = c_{i,k} + \bar{\delta}_{i,k,t} - \underline{\delta}_{i,k,t} + \varrho_{i,t} && \text{(KA1)}\\
d_{m,j,t}: \quad & \lambda_{n(m),t} = u_{m,j} - \bar{\psi}_{m,j,t} + \underline{\psi}_{m,j,t} && \text{(KA2)}\\
f_{\ell,t}: \quad & \kappa_{\ell,t} = \left(A^{\top}\lambda_t\right)_{\ell} + \bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t} && \text{(KA3)}\\
\theta_{n,t}: \quad & \left(A\,\mathrm{diag}(y)\,\kappa_t\right)_{n} + \xi_t \mathbf{1}\{n = r\} = 0 && \text{(KA4)}\\
p_{e,t}: \quad & \lambda_{n(e),t} + \epsilon_e = \alpha_e\left(\pi_{e,t} - \bar{\nu}_{e,t}\right) + \underline{\rho}_{e,t} - \bar{\rho}_{e,t} && \text{(KA5)}\\
g_{e,t}: \quad & \lambda_{n(e),t} = \zeta_e + \beta_e\left(\pi_{e,t} + \underline{\nu}_{e,t}\right) - \underline{\sigma}_{e,t} + \bar{\sigma}_{e,t} && \text{(KA6)}\\
s_{e,t},\ t < T: \quad & \pi_{e,t} = \gamma_e\left(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\right) && \text{(KA7)}\\
s_{e,T}: \quad & \pi_{e,T} = w_e\left(s_{e,T} - \bar{s}_{e,T}\right) && \text{(KA8)}
\end{aligned}
$$

(KA4) gives $\xi = 0$ on left-multiplying by $1^{\top}$, since $1^{\top}A = 0$ — so it reduces to the dual form of Kirchhoff's voltage law.

Dual feasibility and complementary slackness

(KA9)–(KA15)

$$\bar{\mu},\ \underline{\mu},\ \bar{\delta},\ \underline{\delta},\ \bar{\tau},\ \underline{\tau},\ \bar{\psi},\ \underline{\psi},\ \bar{\nu},\ \underline{\nu},\ \bar{\rho},\ \underline{\rho},\ \bar{\sigma},\ \underline{\sigma}\ \geq\ 0 \tag{KA9}$$

$$(\text{A3}):\ 0 \leq \bar{\mu}_{\ell,t}\ \perp\ \bar{f}_{\ell} - f_{\ell,t} \geq 0, \quad 0 \leq \underline{\mu}_{\ell,t}\ \perp\ f_{\ell,t} + \bar{f}_{\ell} \geq 0 \tag{KA10}$$

$$(\text{A5}):\ 0 \leq \underline{\delta}_{i,k,t}\ \perp\ q_{i,k,t} \geq 0, \quad 0 \leq \bar{\delta}_{i,k,t}\ \perp\ \bar{q}_{i,k} - q_{i,k,t} \geq 0 \tag{KA11}$$

$$(\text{A6}):\ 0 \leq \bar{\tau}_{i,t}\ \perp\ \bar{R}_i - Q_{i,t} + Q_{i,t-1} \geq 0, \quad 0 \leq \underline{\tau}_{i,t}\ \perp\ \underline{R}_i - Q_{i,t-1} + Q_{i,t} \geq 0 \tag{KA12}$$

$$(\text{A7}):\ 0 \leq \underline{\psi}_{m,j,t}\ \perp\ d_{m,j,t} \geq 0, \quad 0 \leq \bar{\psi}_{m,j,t}\ \perp\ \bar{d}_{m,j} - d_{m,j,t} \geq 0 \tag{KA13}$$

$$(\text{A9}),\ (\text{A10}):\ 0 \leq \bar{\nu}_{e,t}\ \perp\ \bar{s}_e - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} \geq 0 \tag{KA14}$$

$$0 \leq \underline{\nu}_{e,t}\ \perp\ \gamma_e s_{e,t-1} - \Delta T \beta_e g_{e,t} - \underline{s}_e \geq 0$$

$$(\text{A11}),\ (\text{A12}):\ 0 \leq \underline{\rho}_{e,t}\ \perp\ p_{e,t} \geq 0, \quad 0 \leq \bar{\rho}_{e,t}\ \perp\ \bar{p}_e - p_{e,t} \geq 0 \tag{KA15}$$

$$0 \leq \underline{\sigma}_{e,t}\ \perp\ g_{e,t} \geq 0, \quad 0 \leq \bar{\sigma}_{e,t}\ \perp\ \bar{g}_e - g_{e,t} \geq 0$$

Plus primal feasibility (A1)–(A12). This is the complete optimality system of the ISO's problem — every multiplier belongs to the operator.

2.6 · MODEL B

The multi-agent model: one problem per participant

Each is solved by a price-taking participant that treats λ as given and is subject only to its own constraints. None of the four mentions welfare.

(B1) GENERATOR i

$$\max_{q_{i,\cdot,\cdot}} \ \Delta T \sum_{t\in\mathcal{T}} \Big[\lambda_{n(i),t} Q_{i,t} - \sum_{k} c_{i,k} q_{i,k,t}\Big] \qquad \text{s.t. (A5), (A6)}$$

(B2) LOAD m

$$\max_{d_{m,\cdot,\cdot}} \ \Delta T \sum_{t\in\mathcal{T}} \sum_{j} \big(u_{m,j} - \lambda_{n(m),t}\big) d_{m,j,t} \qquad \text{s.t. (A7)}$$

(B3) STORAGE OWNER e

$$\max_{p_e, g_e, s_e} \ \Delta T \sum_{t\in\mathcal{T}} \Big[\lambda_{n(e),t}\big(g_{e,t} - p_{e,t}\big) - \epsilon_e p_{e,t} - \zeta_e g_{e,t}\Big] + b_e\big(s_{e,T} - \bar{s}_{e,T}\big) \quad \text{s.t. (A8)–(A12)}$$

(B4) NETWORK

$$\max_{f,\theta} \ -\Delta T \sum_{t\in\mathcal{T}} f_t^{\top} A^{\top} \lambda_t \qquad \text{s.t. (A2), (A3), (A4)}$$

(B4) is a modelling device: no merchant transmission firm exists. It is the operator enforcing Kirchhoff, written as an agent so the network block decentralises too.

Model B in full: one square system, no objective

$$\textbf{Find}\ \ z = \big(\underbrace{q,\, d,\, f,\, \theta,\, p,\, g,\, s}_{\text{primal}};\ \underbrace{\lambda,\, \kappa,\, \xi,\, \pi}_{\text{free duals}};\ \underbrace{\bar{\tau},\, \underline{\tau},\, \bar{\nu},\, \underline{\nu}}_{\text{sign-constrained duals}}\ \big) \in \mathcal{B}\ \ \textbf{such that}$$

$$\textit{generator } i\ (\text{B1}): \quad 0 \le q_{i,k,t} \le \bar{q}_{i,k} \ \perp\ c_{i,k} + \varrho_{i,t} - \lambda_{n(i),t} \tag{KB1}$$

$$0 \le \bar{\tau}_{i,t} \ \perp\ \bar{R}_i - Q_{i,t} + Q_{i,t-1} \ge 0, \quad 0 \le \underline{\tau}_{i,t} \ \perp\ \underline{R}_i - Q_{i,t-1} + Q_{i,t} \ge 0$$

$$\textit{load } m\ (\text{B2}): \quad 0 \le d_{m,j,t} \le \bar{d}_{m,j} \ \perp\ \lambda_{n(m),t} - u_{m,j} \tag{KB2}$$

$$\textit{storage } e\ (\text{B3}): \quad 0 \le p_{e,t} \le \bar{p}_e \ \perp\ \lambda_{n(e),t} + \epsilon_e - \alpha_e\big(\pi_{e,t} - \bar{\nu}_{e,t}\big) \tag{KB3}$$

$$0 \le g_{e,t} \le \bar{g}_e \ \perp\ \zeta_e + \beta_e\big(\pi_{e,t} + \underline{\nu}_{e,t}\big) - \lambda_{n(e),t} \tag{KB4}$$

$$\pi_{e,t} - \gamma_e\big(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\big) = 0 \ \ (t < T), \quad \pi_{e,T} - w_e\big(s_{e,T} - \bar{s}_{e,T}\big) = 0 \tag{KB5}$$

$$0 \le \bar{\nu}_{e,t} \ \perp\ \bar{s}_e - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} \ge 0 \tag{KB6}$$

$$0 \le \underline{\nu}_{e,t} \ \perp\ \gamma_e s_{e,t-1} - \Delta T \beta_e g_{e,t} - \underline{s}_e \ge 0$$

$$s_{e,t} - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} + \Delta T \beta_e g_{e,t} = 0 \tag{KB7}$$

$$\textit{network}\ (\text{B4}): \quad -\bar{f}_\ell \le f_{\ell,t} \le \bar{f}_\ell \ \perp\ \kappa_{\ell,t} - \big(A^\top \lambda_t\big)_\ell \tag{KB8}$$

$$\big(A\,\mathrm{diag}(y)\,\kappa_t\big)_n + \xi_t \mathbf{1}\{n = r\} = 0, \quad f_{\ell,t} - y_\ell\big(A^\top \theta_t\big)_\ell = 0, \quad \theta_{r,t} = 0 \tag{KB9}$$

$$\textit{nobody}: \quad \sum_{m \in \mathcal{D}(n),\, j} d_{m,j,t} + D_{n,t} + \sum_\ell A_{n\ell} f_{\ell,t} - \sum_{i \in \mathcal{G}(n),\, k} q_{i,k,t} - \sum_{e \in \mathcal{E}(n)} \big(g_{e,t} - p_{e,t}\big) = 0 \tag{KB10}$$

Over the box $\mathcal{B}$: $0 \le q \le \bar{q}$, $0 \le d \le \bar{d}$, $-\bar{f} \le f \le \bar{f}$, $0 \le p \le \bar{p}$, $0 \le g \le \bar{g}$, $\bar{\tau}, \underline{\tau}, \bar{\nu}, \underline{\nu} \ge 0$, with ϑ, s, λ, κ, ξ, π free.

Three things the box makes visible

No objective

Nothing here is maximised. Social welfare is not a primitive of Model B — it can be computed from a solution, but no row refers to it.

λ is an unknown

It appears in four participants' rows as a datum they take as given, and in (KB10) as the variable that row determines. This is what makes price-taking expressible.

Coupling is λ alone

Delete the λ column and the system falls into $|G| + |D| + |E| + 1$ independent blocks — separability, read as a sparsity pattern.

Every row but the last is some participant's own first-order condition. (KB10) alone belongs to nobody — it is what makes λ consistent with their independent choices.

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Assumptions

What has to hold before the word “equilibrium” can be used

Three hypotheses, doing three different jobs

Assumption 1 · Bid validity

w(e) is non-increasing for every e — equivalently, b(e) is concave. A statement about the curvature of a function.

Assumption 2 · Convex market

No binaries; losses neglected or convexly linearised; offer stacks non-decreasing and demand stacks non-increasing. A statement about the shape of a set.

Assumption 3 · Price taking

Each participant treats λ as given, and the whole horizon is financially binding. A statement about what the participants' problems actually are.

Separability by ownership is a fourth, and independent, hypothesis: it is about the sparsity pattern of the constraint matrix, not about curvature or shape.

Five steps, resting on different hypotheses

step	rests on
KKT necessary for Model A	affine constraints (Assumption 2)
a KKT point exists	Model A feasible, optimum attained
KKT(A) = (KB1)–(KB10)	**separability by ownership**
KKT sufficient per participant	**convexity per agent** (Assumptions 1–2)
each participant globally optimal at λ^*	the definition of equilibrium

Separability buys the partition. Convexity buys the economics. Neither implies the other — which is why they are stated as separate hypotheses.

Existence needs its own argument: an "if and only if" between two empty sets is vacuous. Monotonicity gives a closed convex solution set; non-emptiness needs feasibility and boundedness.

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Conclusion

What is established, and what is not

What is established, and what is not

ESTABLISHED

- The compact model expanded, with every constraint traced to its origin
- Model B stated independently, as an MCP over a box
- KKT(A) is partitioned by the agents' systems — uses separability only
- Under Assumptions 1–3, a solution is a competitive equilibrium
- Individual rationality and revenue adequacy follow

NOT ESTABLISHED

- Truthfulness of w(e): this is an equilibrium of the reported game
- Rolling horizon and real time — the theorem needs the full horizon binding
- Uncertainty; the model is deterministic throughout
- Non-convex extensions: unit commitment, AC network
- Hybrids: the shared interconnection breaks separability

The second column is not a list of failures — it is the scope statement the theorem needs in order to mean anything.

Where this leaves us

The mechanism is understood, the correspondence is proved, and the equilibrium claim now carries its assumptions explicitly. What the design does is move the intertemporal opportunity cost from a forecast the participant must make to a dual the operator computes — and $\pi(e,t)$ is the same number either way.

Open for discussion: truthfulness, rolling horizon, hybrids — and which of them belongs in this paper.